CHANDU PRASAD BHAIRAPU



AI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in production-grade generative AI and agent-based systems, from multi-agent LLM pipelines and RAG to AI evaluation and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by full-stack work with Python and React across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an agent-based GenAI application end-to-end, from requirements to deployment and continuous optimization (ingestion, chunking, prompt engineering, indexing), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction (LangGraph agent orchestration: entity-extraction, relationship-analysis, consolidation, reflection agents in parallel) for complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via systematic evaluation against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer, using AI-assisted development tools (Claude Code, custom skills, agentic coding workflows) with clean code principles for rapid, high-quality development.
- **Tech stack: Python, LLMs, RAG, LangGraph, LangChain, LiteLLM, MCP, React, FastAPI, Azure, Docker, Kubernetes, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph agent orchestration to optimize decision-making across vector database queries, inference, and summarization.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK (infrastructure as code), adhering to clean code practices. Version-controlled via GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, AWS, CDK, EC2, Lambda, Docker, React, FastAPI, GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval against chemist ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation with CDK (IaC).
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), RAG, LangChain, RAGAS, PytrecEval, Hugging Face Transformers, Vector Databases, Azure AI Search, Docker, PyTorch, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Agent Eval Harness with Human Approval | LangGraph | FastAPI | Langfuse | GitHub Actions
- LangGraph agent with human-in-the-loop approval, prompt-injection tests, LLM-as-a-judge scoring and a CI regression gate.

Project: LLM Gateway on AWS | Python | FastAPI | React | TypeScript | Terraform
- LLM API layer on ECS Fargate via Terraform with least-privilege IAM, Secrets Manager and a React/TypeScript cost dashboard.

TECHNICAL SKILLSET

- **Programming:** Python, SQL | **Databases:** PostgreSQL, MongoDB (NoSQL), Neo4j (graph), SPARQL, Vector Databases(Qdrant, Pinecone, Azure AI Search)
- **GenAI & ML:** LLMs, RAG, Agentic Workflow(LangGraph), NLP (LangChain, Hugging Face transformers, NLTK), Machine Learning (Scikit-Learn), Deep Learning (PyTorch, TensorFlow)
- **Cloud & DevOps:** AWS (S3, EC2, Lambda, SQS, DynamoDB, VPC, CDK as infrastructure as code), Azure, Docker, Kubernetes, Git, GitLab, GitLab CI/CD
- **Tools/Frameworks:** FastAPI, React(beginner), Postman, Protege | **Methodologies:** Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles

LANGUAGES

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)